

2025 Standard Specifications for Highway Construction

BRITISH COLUMBIA

Ministry of Transportation and Transit

Volume 2 of 2

Adopted February 1, 2025

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Ministry of Transportation and Transit

2025
STANDARD SPECIFICATIONS FOR HIGHWAY CONSTRUCTION

VOLUME 2

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Construction and Maintenance Branch
Ministry of Transportation and Transit

Cover Photograph provided by TranBC Flickr site
https://www.flickr.com/photos/tranbc/53397215642/in/album-72177720304074161
Nicomen Bridges - Temporary bridge construction after 2021 atmospheric river event
Trans-Canada Hwy #1, approximately 18 km north of Lytton, BC
50° 16' 5.86" N; 121° 24' 4.62" W

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NOTICE TO USERS

Generally, text significantly changed or added since the 2025 edition is shown with single underlining and a vertical bar in the margin; moved text is double underlined; deletions are marked only by the vertical change bars. Not all changes are marked. Minor corrections such as typos have not been marked.

Extensively modified Sections are watermarked as "Revised Section"; new Sections are watermarked "New Section", and neither have updates marked.

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Free PDF versions of the current and historical editions of these specifications are available for downloading at:

https://www2.gov.bc.ca/gov/content/transportation/transportation-infrastructure/engineering-standards-guidelines/standard-specifications-for-highway-construction

Please contact Construction and Maintenance Branch
for questions related to this document.

Construction and Maintenance Branch
Ministry of Transportation and Transit
940 Blanshard Street Level 4C
PO Box 9850 STN PROV GOVT
Victoria BC V8W 9T5
E-mail: RehabMaintenance@gov.bc.ca

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None

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SECTION 635 – ELECTRICAL AND SIGNING
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THE FOLLOWING SECTIONS HAVE BEEN DELETED FROM VOLUME 2:

None.

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SECTION 000

NOTES

000.01 Ministry Names – Within these Standard Specifications, many Ministry names have been genericized, focusing on the relevant area of responsibility rather than the current, full Ministry name. As an example for an issue related to mine safety, the current "Ministry of Mining and Critical Minerals" is referenced as the "Ministry of Mines".

A list of current Ministries may be found at:

https://www2.gov.bc.ca/gov/content/governments/organizational-structure/ministries-organizations/ministries

Where a provincial Act is referenced, the Minister responsible for that Act can be found under the "Acts/Ministers' Responsibilities" link on the BC Laws webpage:

https://www.bclaws.gov.bc.ca/

Referenced publications maintain the name of the Ministry as shown within the document itself.

Upon request by the Contractor, the Ministry Representative will provide clarification as to which Ministry is currently responsible for any particular reference.

000.02 Cross References Abbreviations and Inclusions – To facilitate the cross referencing among the components of each of the Standard Specification, Tender Document Package, the Contract Document Package and the Contract, unless the context by implication requires otherwise, the following is used:

000.02.01 Shorthand Referencing – A reference to XX#.## means to the article, section, subsection, or paragraph #.## of document XX, where "XX" refers to the following, as the case may be:

CT means Conditions of Tender;

GC means General Conditions;

SCT means Invitation to Tender – Supplemental Conditions of Tender;

SGC means Supplemental General Conditions;

SP means Special Provisions; and

SS means Standard Specifications for Highway Construction.

000.02.02 Inclusive of Subclause – A reference to a higher-level clause includes a reference to any subclauses (e.g. XX#.00 includes XX#.01, XX#.02 etc.; XX#.01 includes XX#.01(a), XX#.01(b)(i), etc.).

000.02.03 Statutes – A reference to a statute, whether or not that statute has been defined, means a statute of the Province of British Columbia unless otherwise stated and includes every amendment to it, every regulation made under it and any enactment passed in substitution therefore or in replacement thereof.

000.03 Hyperlinks – In the PDF version of this document, many reference sources have been hyperlinked (such as the reference in SS 000.01 above) and will show as blue underlined text.

Clicking on the link will open the resource in your web browser.

Generally, links to test methods and specification organizations (like CSA or ASTM) will take you to a site where you can purchase and download the associated specifications; a link to a Ministry document, and some others, within the text of a Section should take you to a website where you can download the referenced document free of charge.

Note: Due to MS Word® prioritizing "Track Changes" formatting over other formatting, some newly inserted hyperlinks are not shown in blue. If there is underlined black text that looks like it might warrant a hyperlink, hover your mouse over it to see if it does have an embedded link.

000.03.01 Disclaimer – Any information, including statutes and regulations, that may be found by accessing URLs or websites (including via hyperlinks) in these Standard Specifications are not the official versions and may not be accurate, complete, current or reliable.

000.04 General Condition (GC) References – Where these Standard Specifications make a reference to a General Condition (e.g. GC 38.00 Changes to Work), that reference is to the applicable clause of the Ministry's Major Works General Conditions.

Where the Work is being performed under a different set of terms and conditions (e.g. the Minor Works General Conditions), the reference shall be read as being to the most comparable term and condition of the actual Contract.

In the event there is uncertainty in determining the applicable clause, the Ministry Representative will determine how the reference is to be interpreted.

000.05 Engineer of Record (EoR) / Professional of Record (PoR) – These two terms are used, essentially interchangeably, within these Standard Specifications and are based on the Association of Professional Engineers and Geoscientists of the Province of British Columbia's (APEGBC) use and recent terminology change from EoR to PoR.

Whereas APEGBC is concerned only with their registrants, including Professional Engineers and Geoscientists, usage in these Standard Specifications may also capture other registered professions such as Registered Professional

Foresters (RPF), Registered Professional Biologists (RPBio), and other disciplines, as appropriate to the circumstances.

Each EoR/PoR is professionally responsible to the Ministry for the works they have designed or field-reviewed.

SECTION 604
STEEL TRAFFIC BARRIER CONSTRUCTION

DESCRIPTION

604.01 Scope – This Section covers the construction of roadside and median steel traffic barriers in accordance with the general layout and details indicated on SS Drawings of the SP312 series.

Steel beam guardrail, treated wood posts, steel posts and accessory materials are specified by SS 312.

Design and part number references are taken from the Task Force 13 "Guide to Standardized Roadside Hardware" web resources, as noted in SS 312.

604.02 Provision of Steel Barriers – Barriers of the type(s) called for shall be constructed at the locations and as shown on the Contract Drawings with the materials, accessories and necessary ancillary work all in accordance with the details indicated on the Contract Drawings, Specifications, Special Provisions and/or typical standard SP Drawings or to the direction of the Ministry Representative.

All materials will be supplied by the Contractor, unless otherwise stated in the Special Provisions.

Construction shall be carried out with all labour, tools, equipment and incidentals necessary to complete all barrier work in accordance with good work practice, for a substantially supported and anchored steel barrier developing a continuous beam strength together with necessary crashworthy end treatment.

MATERIALS

604.10 Materials – Detailed material requirements are set out in SS 312 and indicated on SS Drawings SP312-1, SP312-2, SP312-3, SP312-4, SP312-6, SP312-7, SP312-8, SP312-11 and the manufacturer's drawings.

604.11 Treated Wood – Unless otherwise specified, treated wood shall meet the requirements of SS 908.

604.12 Metal Touch-ups – Touch-up treatment for damaged galvanized metal surfaces shall be two heavy coats of zinc-rich paint selected from the Ministry's Recognized Products List under the category of "Additional Paint Coatings – Zinc-Rich Touch-up Paints and Primers".

CONSTRUCTION

604.20 Post Installation – Posts shall be set true to the lines, spacing, depth(s) and height(s) indicated or required to provide a smooth, continuous installation. The layout, type of support and fixing treatment at ends of barriers or where adjacent to abutments and the like shall be as indicated on the Contract Drawings and relevant SS Drawings SP312-1, SP312-2 and SP312-3, and/or to the direction of the Ministry Representative.

The permissible tolerances for post installation are ±1.5 degrees from vertical for plumb and ±6 mm for alignment.

Treated wood posts shall be driven in place with equipment approved by the Ministry Representative or set in augered or dug holes with necessary dampened and well tamped layers of approved backfill material. Excavated material which is unsuitable for use as backfill shall be substituted with granular material at the Contractor's expense. Backfill shall be thoroughly compacted, using pneumatic tampers, in layers not exceeding 150 mm, for the full depth of the excavation.

Driving will be permitted only if no damage results to shoulders and adjacent slopes. In broken rock embankments the prepunching of holes will be permitted only prior to final compaction, surfacing and paving. Construction details appertaining on exposed rock, rock with minimum overburden or on concrete shall be to the indicated requirements and/or the direction of the Ministry Representative.

All work preparatory to and for the installation of posts shall be to the direction and prior approval of the Ministry Representative for each type of sub-surface condition to produce a thoroughly sound guardrail support system.

Surplus excavated material remaining after guardrail support installation shall be disposed of in a uniform manner within the right of way or cleared away, as directed by the Ministry Representative.

Tops of all posts shall be set a uniform 25 mm ±5 mm above the line of the beam's top edge and where necessary cut to line up.

Posts damaged by the driving operation or other damage from their handling and installation not acceptable to the Ministry Representative shall be replaced at the Contractor's expense.

Minor damage to the tops of posts and other wood components and any field cuts and borings shall receive field treatment in accordance with SS 908.

604.21 Steel Beam Erection – Beam sections will normally be supplied or required for joining at 3.81 m intervals.

All punching, cutting or welding shall be shop executed except for special details in unforeseen and exceptional cases and to the prior approval of the Ministry Representative.

Sections shall have full bearing one to another at laps with splicing bolts drawn tight to 100 Nm for a continuous beam effect. Where guardrail is on a curve, the beam sections shall make close contact over the whole splice area. Shop bent beams will normally be supplied or required for curvature radii under 45 metres.

Beam sections shall be bolted to posts so that the edges and centre portion(s) of beams make full bearing with each offset block or where applicable with back-up plate, post or other support.

Bolts shall not extend more than 15 mm beyond the fixing nut when tightened, with any excess removed and the threads burred.

Washers shall be used only where indicated on the SP312 Drawing series.

604.22 End Assemblies – End assemblies shall be carried out strictly according to the manufacturer's specifications unless specifically directed otherwise by the Ministry Representative in writing as to approved alternative methods and the like.

End assemblies shall be "Manual for Assessing Safety Hardware (MASH)" certified.

Note the safety preference for the following items, which will be shown on the Drawings if required:

(a) Continuing guardrail to shield the entire length of any hazard (as indicated on SS Drawing SP312-1), bridge abutment and the like, rather than a bolted connector (like RWE02b).

(b) Approach transitions stiffened by means of reduced post spacing and, where necessary, the use of Thrie-beam or nested twin W-beams.

MEASUREMENT AND PAYMENT

604.90 Measurement and Payment – Measurement of steel traffic barriers will be by the linear metre from end to end of all rail sections along the line of the completed guardrail or centreline of any doubled median barrier and, unless otherwise specified, overall terminal assemblies.

Payment will be made for steel traffic barrier work at the contract unit price(s) for standard steel W-beam shoulder or median barrier guardrail supplied and/or installed complete in place or for Thrie-beam shoulder or median barrier guardrail supplied and/or installed complete in place.

The contract price(s) shall be accepted as full compensation for supply of all materials, labour, tools, equipment and incidental work to complete the required installation including bolting, transitions, curves, all excavation, backfilling, and surplus disposal.

SECTION 635
ELECTRICAL AND SIGNING

PART A – GENERAL

635.01 Scope – This Section covers the requirements for electrical and signing works. Temporary construction signing is covered in SS 194 – Traffic Control.

The installation of materials shall be in accordance with the SS Drawings of the SP635 series unless otherwise specified.

Ministry Electrical Maintenance will be undertaken by non-Ministry forces, referred to as the "Electrical Maintenance Contractor". The Contractor will be required to coordinate the work with the Electrical Maintenance Contractor and the appropriate Ministry Electrical Services Manager in the Regions. Contact information is available from the Ministry Representative, if necessary.

The Contractor shall arrange the supply, installation and removal of temporary C-63 (Traffic Pattern Changed), C-64 (Signal Operation Changed Signs) and W-329 (New) tabs with the Ministry Electrical Maintenance Contractor. The Electrical Maintenance Contractor will supply, install and remove these signs at no expense to the Contractor.

The Contractor shall comply with the following Ministry publications:

(a) Standard Specifications for Highway Construction

(b) Electrical & Traffic Engineering Manual

(c) Electrical & Signing Materials Standards

(d) Manual of Traffic Signs & Pavement Marking

(e) Technical Bulletins & Technical Circulars (in effect at the Tender Closing Date)

Materials used must listed on the Ministry Recognized Products List or approved by the Ministry Representative.

The above documents can be downloaded from the Ministry Engineering website at:

https://www2.gov.bc.ca/gov/content/transportation/transportation-infrastructure/engineering-standards-guidelines

635.02 Work Regulations – All electrical work shall comply with the latest edition of the Canadian Electrical Code and BC amendments, and any applicable standards or regulations published by the Province of British Columbia.

The Contractor shall conform to all applicable Occupational Health and Safety regulations and, if required, submit a WorkSafeBC Notice of Project Form 52E49 before commencing a Ministry project. The Notice of Project form can be completed online at:

https://www.worksafebc.com/en/for-employers/just-for-you/submit-notice-project

If working near overhead power lines the Contractor must submit a WorkSafeBC Assurance of Compliance with Occupational Health and Safety Regulation, Part 19 form 30M33. A sample of this document can be downloaded at:

https://www.worksafebc.com/en/resources/health-safety/forms/assurance-of-compliance-with-occupational-health-and-safety-regulation-part-19-form-30m33

635.03 Electrical Permits & Inspections – Electrical work shall be performed by a contractor licensed to do so by Technical Safety BC. The licensed electrical contractor shall appoint at least one licensed representative whose qualifications comply with the provisions of Technical Safety BC. The Contractor shall provide the Ministry Representative with the name and phone number of the licensed representative prior to starting construction.

Prior to construction the Contractor shall obtain and pay for all permits required by Technical Safety BC and submit a copy of all permits to the Ministry Representative.

Upon completion of an installation and prior to energizing, the Contractor shall advise the Ministry Representative that the work is complete and ready for final inspection. The Ministry Representative will arrange for an inspection of the installation and report deficiencies to the Contractor. The Engineer of Record will conduct a review of the installation and assess conformance to the requirements of the Contract. The Engineer of Record will then report all findings to the Ministry Representative.

Once deficiencies identified by the Ministry Representative and Engineer of Record have been corrected the Contractor will advise the Ministry Representative who will undertake a final inspection and advise if acceptable. If acceptable, the Ministry Representative will advise the Contractor the installation has been completed satisfactorily. If not, additional corrections will be required by the Contractor.

The Ministry will not accept the installation until all work has been approved by Technical Safety BC and the Ministry Representative.

635.04 Electrical Energy Supply – The electrical energy will be supplied from overhead lines of the utility company's secondary distribution system unless otherwise noted on the Drawings.

The Engineer of Record shall determine all service connection points as noted on the Drawings. The Contractor shall confirm all service connection points before installing service equipment.

The Contractor shall be responsible for making the necessary arrangements with the utility company for the connection of new service(s) and, if required, the

disconnection of existing service(s). The Ministry will pay all utility connection costs.

635.05 Materials – The Contractor shall supply all materials necessary for the satisfactory completion of the project except those materials listed in the Special Provisions as supplied by the Ministry. All materials shall be new and conform to the requirements of the Drawings, other Subsections of the Specifications, the Special Provisions, and the Ministry *Electrical and Signing Material Standards*.

An electronic copy of the Electrical and Signing Material Standards may be downloaded free of charge from the Ministry's engineering publications website at:

https://www2.gov.bc.ca/gov/content/transportation/transportation-infrastructure/engineering-standards-guidelines/electrical-engineering

All products contained in the *Electrical and Signing Material Standards* shall be supplied from the Ministry's *Recognized Products List*. This list identifies the manufacturer, the approved product, the product model number, and the product approval date. An electronic copy of the Ministry Recognized Products List can be downloaded from the Ministry's engineering publications website at:

https://www2.gov.bc.ca/gov/content/transportation/transportation-infrastructure/engineering-standards-guidelines/recognized-products-list

This list is updated regularly.

Electrical materials shall meet the requirements of the Canadian Electrical Code, conform to applicable CSA Standards, and meet the approval of Technical Safety BC.

If specified in the Special Provisions, shop drawings shall be submitted to the Ministry Representative for review by the Engineer of Record. Equipment shall not be ordered until the contractor has received approval from the Ministry Representative.

Unless noted otherwise, all permanent signs shall meet current Ministry specifications. The Ministry *Specifications for Standard Highway Sign Materials, Fabrication and Supply* may be downloaded from the Ministry of Transportation engineering website at:

https://www2.gov.bc.ca/gov/content/transportation/transportation-infrastructure/engineering-standards-guidelines/traffic-engineering-safety/traffic-signs-markings

Alternate materials must be approved by the Ministry Representative prior to their use.

635.06 Utilities – Existing utilities are generally not shown on the electrical and signing Drawings. Where utilities are shown on the Drawings, their locations are approximate.

The Contractor shall locate and protect all existing utilities such as power lines, fibre optic cables, telephone lines, gas and oil pipelines, sewers, water works, etc. The Contractor shall conduct its operations in accordance with the requirements of the utility authorities having jurisdiction. In the event of any damage to utilities, the Contractor shall be held responsible for the cost of all necessary repairs and restoration to the satisfaction of the Ministry Representative and Utility Authority.

All costs for locating and protecting utilities will be considered incidental to the Work.

PART B – UNDERGROUND

635.07 Concrete Bases – Concrete bases shall be constructed in accordance with SS Drawings SP635-1.1.1 through SP635-1.1.47 and SP635-1.4.1 through SP635-1.4.5. The use of the term "concrete bases" in this SS 635 shall also mean "concrete spread footings" where applicable.

The Drawings may refer to concrete bases that are poured-in-place or precast. If there is no reference to poured-in-place bases, concrete bases shall be precast.

Concrete single post sign bases listed in the Ministry Recognized Products List (RPL) shall be considered acceptable alternates to the Ministry Standard concrete single post sign bases when approved by the Engineer of Record for the sign base.

Supply and installation of concrete bases shall include the following:

- excavation and backfill;
- supply and installation of concrete and formwork;
- supply and installation of reinforcing steel;
- supply and installation of conduit;
- supply and installation of steel tubing sleeves in sign post bases; and
- installation of anchor bolts.

The Contractor shall check for conflicts with overhead lines prior to excavating for concrete bases. If it appears there may be an overhead conflict, the Contractor shall contact the Ministry Representative for further instructions. If the Contractor installs a concrete base in a location where the pole conflicts with overhead power lines, the Contractor shall remove the pole and relocate the concrete base to a location approved by the Ministry Representative. The Contractor shall pay all costs for pole and base relocation.

635.07.01 Excavation and Backfill – Excavations shall meet the requirements, but not the Quantities and Payment provisions of SS 407 – Foundation Excavation.

All excavation work shall be carried out as required to suit concrete bases. Where directed by the Ministry Representative, excavations shall be shored to avoid the cutting of pavement, sidewalks, and curb and gutter.

Base excavations shall be backfilled using the excavated material provided it consists of clean well graded granular soil having a maximum fines content of 8% (silt and clay size particles) and a maximum aggregate size of 100 mm. Where backfill material does not conform as noted above, base excavation shall be backfilled with 25 mm Well Graded Base Course Aggregate material as specified in SS 635.16. Unacceptable materials shall be disposed of as specified in SS 635.17.

Backfill shall be placed in layers not exceeding 150 mm compacted thickness (100 mm compacted thickness in the top 300 mm) and shall be compacted to a minimum 100% of the Standard Proctor maximum dry density obtained by ASTM D698. Layer thickness shall be reduced and moisture content of the material adjusted as required to achieve compaction. Care shall be taken not to damage conduits.

All excavated and backfilled areas shall be restored to their original condition.

Asphalt restoration shall be performed in accordance with SS 635.12.

Concrete curb and gutter restoration shall be performed in accordance with SS 635.15.

635.07.02 Concrete and Formwork – Precast concrete bases shall meet the requirements of CSA A23.4. Suppliers of precast concrete bases listed on the Ministry Recognized Products List shall be considered as prequalified manufacturers of concrete bases under CSA A23.4.

Alternatively, manufacturers shall be prequalified under CSA A23.4 by one of the following certification bodies:

(a) The Canadian Precast Concrete Quality Assurance (CPCQA) Certification Program

(b) The CSA Group Testing & Certification Inc. (CSA) Certification Program.

(c) QCAST Certification Program by the American Concrete Pipe Association for Precast Products

(d) National Precast Concrete Association (NPCA) Plant Certification

(e) Other certification organizations acceptable to the Ministry Representative that are accredited by the Standards Council of Canada for the certification of manufacturers for these products.

Manufactures shall be prequalified under CSA A23.4 by one of the above listed certification bodies.

Cast-in-place concrete bases and precast concrete bases that are fabricated on site shall meet the requirements of CSA A23.1 and the Contractor shall notify the Ministry Representative 48 hours prior to pouring concrete. Proportioning of the concrete mixes shall be the responsibility of the Contractor. SS 211 shall only apply when specifically referenced.

The concrete mix shall meet the specifications listed in Table 635-A.

Table 635-A: Concrete for Concrete Bases(1)

Property	Criteria
Minimum compressive strength at 28 days	30 MPa
Nominal maximum aggregate size	20 mm or 28 mm
Maximum w/cm ratio by mass	0.40
Air content – 20 mm nominal maximum aggregate	5-8%
Air content – 28 mm nominal maximum aggregate	4-7%
Slump	120 ± 30 mm
Curing Type (CSA A23.1 or CSA A23.4)	In accordance with class of exposure C-1
Chloride ion penetrability(2) (ASTM C1202 or CSA A23.2-23C)	less than 1,500 coulombs at 91 days

(1) Concrete shall also meet the requirements for S-3 exposure unless otherwise verified by soil test

(2) Reinforced concrete only

All exposed concrete surfaces shall be given a Class 3 finish and all buried surfaces a Class 1 finish in accordance with SS 211.17.

When constructing poured-in-place concrete bases, the Contractor shall have the concrete compressive strength verified, using concrete test cylinders, prior to installing the structure onto the base.

The Contractor must provide written confirmation of concrete test cylinder compressive strength results prior to installing any structure on the bases.

The top finished surface of a concrete base shall not vary by more than 3 mm in depth as measured across the widest surface. If this variation exceeds 3 mm then the top 50 mm of the base shall be broken off by hand and reformed or the base shall be replaced, as directed by the Ministry Representative.

635.07.03 Reinforcing Steel – Reinforcing steel shall meet the requirements, but not the payment provisions, of SS 412 – Reinforcing Steel.

All reinforcing steel shall conform to CSA G30.18-M, Grade 400W or 400R.

The Contractor shall adjust the spacing of reinforcing steel to suit anchor bolts and conduit.

635.07.04 Anchor Bolts and Conduit – Anchor bolts and conduits shall be cast into the concrete bases.

Where an anchor bolt is damaged it shall be repaired in accordance with SS Drawings SP635-1.1.43 and SP635-1.1.44.

Welding of anchor bolts shall not be permitted.

635.07.05 Steel Pipe Sleeves – Steel pipe sleeves shall be cast into the concrete bases. The type of sleeve shall be as noted on the Drawings and shall be in accordance with SS 635.30 or SS 635.31.

635.07.06 Payment – Payment for CONCRETE BASES will be at the Unit Price per base.

The Unit Price shall include all costs for the supply and installation of the bases, including but not limited to, all Type D excavation; supply and installation of shoring to meet Workers Compensation Act and Occupational Health and Safety Regulation requirements; formwork, concrete, reinforcing steel and conduit; supply and installation of grout and sand to fill knock-out voids in bases, controller bases and concrete pads in front of controller bases; supply and installation of steel sleeves in sign post bases; installation of individual anchor bolts or anchor bolts in cages; placing and compaction of excavated material as backfill; removal and disposal of excess excavated material; and all other labour, equipment, and materials necessary to complete the installation.

Any additional costs for supplying higher strength concrete shall be borne by the Contractor.

Payment for excavation of Type A material (solid rock) and de-watering of excavations will be made in accordance with GC 38.00 Changes to Work.

Where excavated material is ruled unacceptable for backfill, payment for 25 mm Well Graded Base Course Aggregate used as backfill will be made under SS 635.16.01.

635.08 Junction Boxes and Vaults

635.08.01 General

(a) Plastic Junction Boxes – Plastic junction boxes and their lids/hatches shall be in accordance with SS Drawings SP635-1.2.1 through SP635-1.2.16 and SP635-1.4.1 through SP635-1.4.3.

(b) Concrete Junction Boxes – Concrete junction boxes and their lids/hatches shall be in accordance with SS Drawings SP635-1.3.1 through SP635-1.3.4.

(c) Concrete Vaults – Concrete vaults and their lids/hatches shall be in accordance with SS Drawings SP635-1.3.3 and SP635-1.3.4.

635.08.02 Plastic Junction Boxes – Plastic junction boxes listed in the Contract documents, Special Provisions or in the Ministry Recognized Products List (RPL) shall be considered acceptable. Installation of plastic junction boxes shall be accordance with SS Drawings SP635-1.2.1 through SP635-1.2.16 and SP635-1.4.1 through SP635-1.4.3.

635.08.03 Concrete Junction Boxes and Vaults Not Located in an Area with Routine Vehicle Access – Concrete junction boxes, concrete vaults and their lids/hatches that are located in areas that do not have routine traffic access, including but not limited to, sidewalks and landscaped areas, shall meet the requirements of this SS 635.08.03.

Concrete junction boxes, concrete vaults and their lids/hatches that are located in areas that do not have routine traffic access shall be designed by engineers registered with the Association of Professional Engineers and Geoscientists of the Province of BC. The design shall be in accordance with recognized and applicable CSA, ASTM or AASHTO standards. Design drawings and Letters of Assurance meeting the requirements of the Ministry Technical Circular T-01/23 Professional Assurance Guidelines shall be submitted to the Ministry Representative for review and acceptance at least two (2) weeks prior to delivery. Concrete vaults and their lids/hatches shall meet the requirements for non-deliberate vehicular loading using the wheel loading for either of the following:

(a) The design live load in accordance with the Ministry Supplement to the CHBDC S6, or

(b) AASHTO Standard Specification for Highway Bridges HL-93, H-20 or HS-20

635.08.04 Concrete Junction Boxes and Vaults located in Traffic Areas – Concrete junction boxes, concrete vaults and their lids/hatches that are located in areas with routine traffic access, including but not limited to, vehicular roadways, vehicular pull-outs, parking areas, and driveways, shall meet the requirements of SS 635.08.04.

Concrete junction boxes, concrete vaults and their lids/hatches shall be designed by engineers registered with the Association of Professional Engineers and Geoscientists of the Province of BC. Design drawings and Letters of Assurance meeting the requirements of the Ministry Technical Circular T-01/23 Professional Assurance Guidelines shall be submitted to the Ministry Representative for review and acceptance at least two (2) weeks prior to delivery.

Concrete junction boxes and concrete vaults located in areas with routine vehicle access shall be designed in accordance with the Canadian Highway Bridge Design Code (CHBDC) CSA S6 and the Ministry Supplement to CHBDC S6. The design live load shall be in accordance with the Ministry Supplement to the CHBDC S6.

The lids/hatches for concrete vaults that are located in areas with routine traffic access shall be designed in accordance with either of the following:

(a) Canadian Highway Bridge Design Code (CHBDC) CSA S6 and the Ministry Supplement to CHBDC S6, or

(b) ASTM 1802 – Standard Specification for Design, Testing, Manufacture, Selection, and Installation of Fabricated Metal Horizontal Access Hatches for Utility, Water, and Wastewater Structures, Load Level 7, Full Traffic.

635.08.05 Concrete for Junction Boxes and Vaults – SS 211 shall only apply to concrete for junction boxes and vaults when specifically referenced.

(a) Concrete for junction boxes and vaults that are located in areas that do not have routine traffic access shall be in accordance with CSA A23.1 and CSA A23.4 based on a minimum service life of 40 years and shall have minimum cover of 40 mm on exterior surfaces and 25 mm on interior surfaces. The concrete mix shall meet the specifications listed in Table 635-B.

(b) No-slump concrete meeting the requirements of CSA A23.1 shall be an acceptable alternative for concrete junction boxes located in areas that do not have routine traffic access. Galvanized or stainless steel reinforcement shall be used with no-slump concrete when the minimum concrete cover identified above cannot be achieved for thin products. When using galvanized or stainless steel reinforcement with no-slump concrete, the minimum product thickness shall be 25 mm and the minimum cover shall be 10 mm. When requested, the Contractor shall submit the proposed mix design and detailed documentation for ministry review, a minimum of 14 days in advance of fabrication, which shall include as a minimum the following:

(i) Details of the products the Contractor proposes to fabricate using no-slump concrete, and

(ii) Reinforcement details and concrete cover for reinforcement, and

(iii) Concrete mix design submittals in accordance with SS 635.09, and

(iv) Documentation demonstrating that the proposed mix design will achieve the required strength, durability, and performance requirements including the documentation and test results for the proposed mix design in accordance with CSA A23.1 Clause 8.9 – No-slump concrete.

The Ministry Representative may also request additional documentation be submitted to assist in the ministry review of the mix design for no-slump concrete. If, in the sole discretion of the Ministry Representative, the submitted documentation does not demonstrate acceptable performance of the no-slump concrete, then no-slump concrete shall not be used.

(c) Concrete for junction boxes and vaults that are located in areas with routine traffic access shall be in accordance with CSA S6 for exposure to harsh conditions. The concrete mix shall meet the specifications listed in Table 635-B.

Table 635-B: Concrete for Junction Boxes and Vaults(1)

Property	Criteria
Minimum compressive strength at 28 days	35 MPa
Air content category as per CSA A23.1 Table 4	1
Maximum w/cm ratio by mass	0.40
Curing Type (CSA A23.1 or CSA A23.4)	In accordance with class of exposure C-XL
Chloride ion penetrability (ASTM C1202 or CSA A23.2-23C)	less than 1,000 coulombs at 91 days

(1) (1) Concrete shall also meet the requirements for S-3 exposure unless otherwise verified by soil test.

635.08.06 Concrete Junction Box and Vault Fabrication

(a) Certified Manufacturers – For all contracts awarded after December 31, 2021, concrete junction boxes and vaults shall be fabricated in facilities that are certified by an independent industry recognized certification body specific to the products being fabricated.

The certification body shall be qualified to inspect and evaluate all the relevant procedures, facilities, and personnel of the manufacturer as to conformity with required standards and shall be acceptable to the Ministry. Certification shall be in effect prior to the beginning of Work and shall be maintained throughout the period of manufacture.

Concrete junction boxes and vaults shall be fabricated in accordance with CSA A23.4. Acceptable certification bodies for concrete junction boxes and vault manufacturers include:

(i) The Canadian Precast Concrete Quality Assurance (CPCQA) Certification Program

(ii) Certification organizations accredited by the Standards Council of Canada (i.e. Canadian Standards Association (CSA))

(iii) QCAST Certification Program by the American Concrete Pipe Association for Precast Products

(iv) National Precast Concrete Association (NPCA) Plant Certification

(b) Lids and Hatches – All junction box and vault lids and hatches shall be galvanized steel and shall be fabricated in accordance with SS 422.

635.08.07 Installation – Excavations for junction boxes and vaults shall meet the requirements, but not the Quantities and Payment provisions, of SS 407 – Foundation Excavation.

All excavation work shall be carried out as required to suit junction boxes and vaults. Where directed by the Ministry Representative, excavations shall be shored to avoid the cutting of pavement, sidewalks and curb and gutter.

Junction boxes and vaults shall be installed on a bedding of 25 mm Well Graded Base Course Aggregate material in accordance with SS 202 – Granular Surfacing, Base and Sub-bases.

If accepted by the Ministry Representative, junction box or vault excavations may be backfilled using the excavated material. Where new material is required, 25 mm Well Graded Base Course Aggregate material shall be used as specified in SS 635.16.

Unacceptable materials shall be disposed of as specified in SS 635.17.

Bedding and backfill material shall be placed in layers not exceeding 100 mm compacted thickness and shall be compacted to a minimum 95% of the Standard Proctor maximum dry density obtained by ASTM D698. Layer thickness shall be reduced and moisture content of the material adjusted as required to achieve compaction. Care shall be taken not to damage conduits.

Unless otherwise shown, the top of junction boxes and vaults shall be installed so that they are flush with the finished grade. Concrete for junction box collars shall conform to SS 635.08.05.

All areas where excavation and backfilling have been performed shall be restored to their original condition.

Asphalt restoration shall be performed in accordance with SS 635.14.

Concrete curb and gutter restoration shall be performed in accordance with SS 635.15.

Lid/hatch hold down bolts shall be coated with anti-seize lubricant.

635.08.08 Payment – Payment for PLASTIC JUNCTION BOXES will be at the Contract Unit Price per junction box.

The Unit Price shall include all costs of excavation, other than concrete and asphalt removal; installation of junction box sections, steel lids/hatches, drain plates, adapter plates and braces; supply and installation of shoring to meet WCB requirements, conductor support bars in boxes, concrete collars, 25 mm Well Graded Base Course Aggregate material below the junction box; placing and compaction of excavated material as backfill; removal and disposal of excess excavated material; and all other labour, equipment and materials necessary to complete the installation.

Payment for the excavation of Type A material (solid rock), de-watering of excavations, and additional drainage in vaults will be made in accordance with GC 38.00 Changes to Works.

Where excavated material is ruled unacceptable for backfill, payment for 25 mm Well Graded Base Course Aggregate will be made under SS 635.16.01.

635.08.09 Payment – Payment for CONCRETE JUNCTION BOXES AND VAULTS will be at the Contract Unit Price per junction box or vault.

The Unit Price shall include all costs of excavation, other than concrete and asphalt removal; supply and installation of the concrete junction box or vault complete with collar and lid/hatch; supply and installation of conductor supports; supply and installation of 25 mm Well Graded Base Course Aggregate material below concrete junction box or the vault; placing and compaction of excavated material as backfill; removal and disposal of excess excavated material; and all other labour, equipment and materials necessary to complete the installation.

Payment for the excavation of Type A material (solid rock) de-watering of excavations, and additional drainage in vaults will be made in accordance with GC 38.00 Changes to Works.

Where excavated material is ruled unacceptable for backfill, payment for 25 mm Well Graded Base Course Aggregate will be made under SS 635.16.01.

635.09 Concrete Mix Design Submittals – When requested by the Ministry Representative, the Contractor shall submit concrete mix design information for bases, junction boxes and vaults for Ministry quality assurance and compliance review. Submittals shall be done using the requirements of CSA A23.1 Table 5 Alternative 1. The submittals shall meet the requirements of CSA A23.1 and include the following.

635.09.01 Project Specifics

(a) Name and location of the proposed supplier.

(b) Expected method of batching, transporting and placing concrete.

(c) Mix parameters including plastic properties, structural requirements and durability requirements.

(d) Name and contact information of quality control testing laboratory and certified testing personnel.

635.09.02 Materials

(a) Cementitious Materials – Types and source of each material including mill test reports and manufacturer's certificates of compliance.

(b) Aggregates – Type and source of all individual aggregate products including individual gradations and aggregate quality testing

(c) Admixtures – Type and source of all admixtures, point of addition to the mix, and individual technical data sheets.

(d) Water – The source of mixing water and water quality test data

(e) Miscellaneous – Documentation for all other materials proposed for the mix, showing conformance with applicable requirements and manufacturer/industry guidelines and standards.

635.09.03 Documentation – Documentation establishing that the proposed mix design will achieve the required plastic properties, strength, durability, and performance requirements.

635.10 Self Consolidating Concrete – Self Consolidating Concrete (SCC) meeting the requirements of CSA A23.1 may be considered for use on a product and project specific basis. The use of SCC is subject to the approval of the Ministry Representative.

635.10.01 Consideration of SCC – For the use of SCC to be considered, the Contractor shall submit the proposed mix design and detailed documentation for ministry review, a minimum of 28 days in advance of fabrication, which shall include as a minimum the following:

(a) Details of the products the Contractor proposes to fabricate using SCC, and

(b) Concrete mix design submittals in accordance with SS635.09, and

(c) Documentation in accordance with either of the following:

(i) Details in accordance with CSA A23.2-24C documenting the proven history of successful performance for the required strength, durability, and other performance requirements of the proposed mix design in the fabrication of products similar to those proposed for the use of SCC in the Work, or

(ii) A full-scale test shall be used to verify the self-consolidating characteristics for placement and for the hardened concrete properties of the mix design for the proposed application. Documentation in accordance with CSA A23.2-24C shall be submitted demonstrating that the proposed mix design will achieve the required strength, durability, and performance requirements.

The Ministry Representative may also request additional documentation be submitted to assist in the ministry review. If, in the sole discretion of the Ministry Representative, the submitted documentation does not demonstrate acceptable performance of the SCC, then SCC shall not be used.

635.11 Conduits – All electrical conduits shall be installed in accordance with the Drawings.

Rigid PVC (RPVC) conduit shall be unplasticized polyvinyl chloride and conform to CSA C22.2 No. 211.2. Couplings, adapters, bends and fittings shall be unplasticized polyvinyl chloride and conform to CSA C22.2 No. 85. RPVC conduit shall be installed using CSA certified cement. Each standard length of RPVC conduit and fitting shall bear a certification mark to the applicable CSA standard.

Rigid metal conduit (RMC) shall be hot-dipped galvanized rigid steel and conform to CSA C22.2 No. 45. Any exposed metal on conduit shall be coated with cold galvanizing compound in accordance with SS 635.25.

High-density polyethylene (HDPE) conduit shall conform to CSA 22.2 No. 327. Markings on the conduit shall meet the requirements of the Canadian Electrical Code. Communication conduits shall be orange in colour.

Flexible liquid-tight non-metallic conduit (FC) shall conform to CSA 22.2 No. 227.2. All joints shall be made with FC threaded couplers, adapters or conduit fittings.

Flexible liquid-tight metal conduit (FMC) shall conform to CSA 22.2 No. 56. All joints shall be made with FMC threaded couplers, adapters or conduit fittings.

Conduit straps shall be galvanized steel single hole or double hole type (sized to suit conduit).

All empty conduits shall have a pull string and shall be capped in accordance with SS Drawings SP635-1.2.16, SP635-1.3.5, and SP635-1.7.1. Pull string shall be polypropylene with a minimum tensile strength of 1.1 kN.

The Contractor shall lay out conduit so that 90° bends are avoided. Where 90° bend cannot be avoided, only one will be allowed for each conduit run.

Conduit couplings or bell ends damaged when pulling conductors shall be rejected and replaced at the Contractor's expense. An appropriate pulling jig or method shall be used to avoid damaging conduit or fittings when pulling conductors.

635.11.01 Payment – Payment for CONDUIT will be at the Unit Price per metre.

The Unit Price shall include all costs for the supply and installation of underground conduit, fittings, cement and pull strings; and all other labour, equipment and materials necessary to complete the installation.

Payment for conduits installed above ground on poles or sign structures is not included in this Subsection.

635.12 Trenching and Backfilling for Conduit – Trenching and backfilling shall be performed in accordance

with SS Drawings SP635-1.5.1, SP635-1.5.2, SP635-1.6.1, SP635-1.6.2 and SP635-1.7.1.

RPVC conduits shall be installed in an open trench in accordance with SS Drawings SP635-1.5.1, SP635-1.5.2, SP635-1.6.1, SP635-1.6.2 and SP635-1.7.1 except where conduits are to be drilled or pushed under the surface as specified in SS 635.13.

Trenches shall be excavated with neat, uniform sides to the minimum width necessary, but not less than the minimum dimensions indicated on SS Drawings SP635-1.5.1 and SP635-1.5.2 (care shall be taken to prevent sloughing). Material in the floor of the trench shall be undisturbed, or if disturbed shall be re-compacted to the requirement herein for backfill.

Where approved by the Ministry Representative excavated material may be used as backfill. Where new material is required, 25 mm Well Graded Base Course Aggregate material shall be used as specified in SS 635.16. Unacceptable materials shall be disposed of as specified in SS 635.17.

A 150 mm wide yellow plastic marker tape indicating "WARNING ELECTRICAL" shall be installed in each trench for the complete length of the trench. The location of the marker tape shall be as shown on SS Drawings SP635-1.5.1 and SP635-1.5.2.

Bedding and backfill material shall be placed in layers not exceeding 100 mm compacted thickness and shall be compacted to a minimum 100% (for road crossings) and 95% (for shoulder trenches) of the Standard Proctor maximum dry density obtained by ASTM D698. Layer thickness shall be reduced, and moisture content of the material adjusted as required to achieve compaction. Care shall be taken not to damage conduits.

All areas where trenching and backfilling have been performed shall be restored to their original condition.

Asphalt restoration shall be performed in accordance with SS 635.14.

Concrete curb and gutter and sidewalk restoration shall be performed in accordance with SS 635.15.

635.12.01 Payment – Payment for TRENCHING AND BACKFILLING FOR CONDUIT will be at the Unit Price per metre.

The Unit Price shall include all costs of Type D excavation, other than concrete and asphalt removal; placing and compaction of excavated material as backfill; dewatering; supply and installation of trench marker tape; removal and disposal of excess excavated material; and all other labour, equipment and materials necessary to complete the installation.

Payment for excavation of Type A material (solid rock) will be made in accordance with GC 38.00 Changes to Work.

Where excavated material is ruled unacceptable for backfill, payment for 25 mm Well Graded Base Course Aggregate will be made under SS 635.16.01.

635.13 Trenchless Conduit Installation – Where noted on the Drawings the Contractor shall install the conduit using a suitable trenchless technology to avoid cutting the existing pavement or sidewalk.

The trenchless technology selected shall suit the site conditions.

The Contractor shall notify the Ministry Representative a minimum of 48-hours prior to installing conduits.

The Contractor shall locate all underground utilities prior to installing conduits.

635.13.01 Payment – Payment for installing conduit using trenchless technology will be at the Contract Unit Price per metre.

The Unit Price shall include all costs of excavation for set up; backfilling and restoration when finished, and all other labour, equipment and materials necessary to complete the installation.

No separate payment will be made for unsuccessful installation.

Payment for the supply and installation of conduit is covered under SS 635.11.01.

635.14 Asphalt Pavement – Asphalt paving shall be performed in accordance with SS 502.

Asphalt patching of conduit trenches shall be performed in accordance with SS Drawing SP635-1.5.1. Asphalt patching of excavations for junction boxes, vaults and bases shall be performed similar to the details shown on SS Drawing SP635-1.5.1.

Contrary to the requirements of Section 502, Asphalt placing by a paving machine will not be required. Asphalt pavement restoration shall be Class 1 medium mix installed in three 50 mm lifts unless otherwise noted or directed by the Ministry Representative.

The Contractor shall use a pavement saw to cut pavement and shall exercise care to obtain a neat, uniform excavation line along the edge of the pavement. The Contractor shall also prevent the existing base gravel from sloughing and undermining the pavement.

Where three lifts of asphalt pavement are being placed to reinstate a trench, the first two lifts shall be compacted with vibratory equipment capable of operating within the excavated area and of compacting over the whole of it. The third lift shall be compacted with rolling equipment to produce a uniform surface matching the adjacent existing pavement.

635.14.01 Payment – Payment for ASPHALT PAVEMENT for electrical and signing work will be as follows.

(a) Where the electrical work forms part of a paving or road construction Project, measurement and payment will be made in accordance with SS 502.

(b) Where the electrical work does not form part of a paving or road construction Project, payment shall be made in accordance with SS 502, except that:

(i) No separate payment shall be made for spray primer and tack coat and all costs of this work shall be included.

(ii) Measurement for payment for conduit trenches shall be made in cubic meters of mix compacted in place and shall be calculated using a 450 mm wide trench plus 300 mm (asphalt is excavated 150 mm beyond the trench on both sides as noted on SS Drawing SP635-1.5.1).

(iii) Measurement for payment for installation of all other works shall be made in cubic metres of mix compacted in place and shall be calculated from measurements of the work.

635.14.02 Payment – Payment for REMOVAL OF ASPHALT will be made at the Contract Unit Price per cubic metre.

The Unit Price shall include all costs of cutting, removing and disposing of existing asphalt; and all other labour, equipment and materials necessary to complete the removal of asphalt.

The volume for payment of asphalt removal for conduit trenches will be calculated by the actual volume of asphalt removed within a 450 mm wide trench plus 300 mm (asphalt is excavated 150 mm beyond the trench on both sides as noted on SS Drawing SP635-1.5.1).

The volume for payment of asphalt removal for all other work will be calculated from measurements of the work.

635.15 Concrete Curbs and Sidewalks – Concrete curbs and sidewalks shall be installed in accordance with SS 582 – Concrete Curb and Gutter and Storm Drainage.

Concrete shall meet the requirements of Note 1 on SS Drawing SP582-01.01.

The Contractor shall use a pavement saw to cut existing concrete unless otherwise directed by the Ministry Representative, and shall obtain a neat, uniform cut along the edge of concrete. The Contractor shall also prevent base gravel from sloughing and undermining the concrete.

635.15.01 Payment – Payment for CONCRETE CURBS will be at the Contract Unit Price in accordance with SS 582 – Concrete Curb and Gutter and Storm Drainage.

The Unit Price shall include all costs of excavation, other than concrete and asphalt removal; supply and installation of 25 mm Well Graded Base Course Aggregate bedding material; supply and installation of formwork, concrete and expansion joint material; removal and disposal of excess excavated material; and all other labour, equipment and materials necessary to complete the installation.

635.15.02 Payment – Payment for SIDEWALKS will be at the Contract Unit Price per square metre.

The Unit Price shall include all costs of excavation, other than concrete and asphalt removal; supply and installation of 25 mm Well Graded Base Course Aggregate bedding material; supply and installation of formwork, concrete and expansion joint material; removal and disposal of excess excavated material; and all other labour, equipment and materials necessary to complete the installation.

635.15.03 Payment – Payment for REMOVAL OF CONCRETE will be at the Contract Unit Price per cubic metre.

The Unit Price shall include all costs for the cutting, removing and disposing of existing concrete roadway, sidewalk, etc.; and all other labour, equipment and materials necessary to complete the removal of concrete.

The volume for payment of concrete removal for conduit trenches will be calculated by the actual volume of concrete removed within a 450 mm wide trench.

The volume for payment of concrete removal for all other work will be calculated from measurements of the work.

635.15.04 Payment – Payment for REMOVAL OF CONCRETE CURBS will be at the Contract Unit Price per metre.

The Unit Price shall include all costs for cutting, removing and disposing of existing concrete; and all other labour, equipment and materials necessary to complete the removal of concrete curbs.

635.16 25 mm Well Graded Base Course Aggregate – 25 mm Well Graded Base Course Aggregate shall meet the requirements of SS 202 – Granular Surfacing, Base and Sub-bases.

When acceptable excavated material is not available, 25 mm Well Graded Base Course Aggregate shall be used as backfill. It shall also be used for additional backfill around bases, junction boxes, vaults, median island fill, and other work as approved by the Ministry Representative.

635.16.01 Payment – Payment for 25 mm WELL GRADED BASE COURSE AGGREGATE will be at the Contract Unit Price per cubic metre.

The Unit Price shall include all costs of the supply, hauling, placing, trimming and compacting the material, and all other labour and equipment necessary to complete the installation.

Where the 25 mm Well Graded Base Course Aggregate material is used to backfill a conduit trench, its volume shall be calculated in place using:

(a) The actual length of trench from which the material is designated unacceptable;

(b) a 450 mm wide trench; and

(c) the actual depth of this material compacted in place.

Where 25 mm Well Graded Base Course Aggregate material is used to backfill bases, junction boxes, vaults, or other work the volume will be calculated by the actual compacted volume of material installed in the excavation less the volume of the item installed in the excavation.

635.17 Removal of Rejected Excavated Material – Where excavated material is designated unacceptable for backfill by the Ministry Representative it shall be removed from the work site and disposed of by the Contractor.

The disposal site shall be approved by the Ministry Representative.

635.17.01 Payment – Payment for the REMOVAL OF REJECTED EXCAVATED MATERIAL will be at the Contract Unit Price per cubic metre.

The Unit Price shall include all costs of removal and disposal, and all other labour, equipment and materials necessary to complete the removal of rejected material.

Where the rejected material has been excavated from a conduit trench, its volume shall be calculated in place using:

(a) The actual length of trench from which the material is designated unacceptable;

(b) a 450 mm wide trench; and

(c) the actual depth of material removed.

Where the rejected material has been excavated for bases, junction boxes, vaults or other work the volume will be calculated by the actual volume of excavated material less the volume of the item installed in the excavation.

635.18 Restoration – Areas where work has been performed shall be returned to their original condition and must be left in a neat state to the satisfaction of the Ministry Representative. All costs for restoration will be considered incidental to the Work.

PART C – ELECTRICAL

635.19 Luminaire, Signal and Sign Poles – Luminaire and signal poles shall be installed in accordance with SS Drawings SP635-2.1.1 through SP635-2.1.14, and SP635-2.2.1 through SP635-2.2.10.

Where specified, luminaire poles shall be mounted on frangible or breakaway bases in accordance with SS Drawings SP635-2.1.15 and SP635-2.1.16.

Sign poles shall be installed in accordance with SS Drawings SP635-3.1.1 through SP635-3.1.22.

Poles shall not be erected until concrete bases have attained a strength of 30 MPa.

Push button and signal head locations shall be verified with the Drawings, before poles are drilled and assembled.

Field drilling of holes larger than 33 mm (1 5/16") diameter will not be permitted in Type 1, 3, 6, 7, L, M and H shafts, and all arms and extensions. Where larger holes are required, the holes shall be reinforced with a welded bushing in accordance with SS Drawing SP635-2.4.17.

All components of the luminaire, signal and sign poles shall be handled with care to prevent stress to the components through bending or twisting. A nylon sling shall be used to transport and erect the components. The use of steel chains as slings will not be permitted. Any damage to the components through overstress, scratching or denting shall be repaired or replaced at the Contractor's expense to the satisfaction of the Ministry Representative.

All pole shafts shall be installed plumb. Where possible, all luminaire poles shall be positioned with the hand-hole oriented opposite the road and if not possible, on the downstream traffic side.

All wiring inside the poles shall be in accordance with SS 635.21.

Pole shafts shall mount directly on concrete bases and, where necessary, shall be trued to plumb using levelling shims.

The Contractor shall tighten all bolts and nuts to 1/3 rotation past snug tight. "Snug-tight" is the tightness attained by a few impacts of an impact wrench or the full effort of a person using a spud wrench.

All scratches in poles and all field drilled holes shall be coated with cold galvanizing compound as per SS 635.25.

Hand hole bolts shall be coated with anti-seize lubricant.

635.19.01 Traffic and Pedestrian Signal Heads – Traffic and pedestrian signal heads shall be installed in accordance with SS Drawings SP635-2.3.1 through SP635-2.3.7 and SP635-2.3.10.

All traffic and pedestrian signal heads and mounting hardware shall be securely attached to the pole.

The Contractor shall aim all traffic signal heads in accordance with SS Drawing SP635-2.3.11. Pedestrian signal heads shall be aimed to line up with the opposing wheelchair ramp or centre of the crosswalk.

Traffic and pedestrian signal heads shall be aimed in the field to the satisfaction of the Ministry Representative.

All traffic and pedestrian signal heads shall have LED (light emitting diode) traffic signal heads unless noted otherwise on the Drawings.

The Contractor shall completely cover all traffic and pedestrian signal heads with dark coloured pre-manufactured signal cover bags from the time they are installed until the system is required by the Ministry.

635.19.02 Pedestrian Pushbuttons and Signs – Pedestrian pushbuttons and signs shall be installed in accordance with SS Drawing SP635-2.3.8 and SP635-2.3.14.

Pedestrian push buttons and signs shall be securely attached to the pole. Pedestrian pushbutton signs shall be temporarily covered until the signal is in operation.

635.19.03 Luminaires and Photocells – Luminaires and photocells shall be installed as per manufacturer's instructions or as noted on the Drawings. Luminaires shall be installed level.

Roadway luminaires shall be oriented parallel to the roadway surface to reduce glare.

Sign luminaires are typically no longer used. If specified, sign luminaires will require different aiming depending on the manufacturer. Aiming angles shall be noted on the Drawings or shall be requested from the Ministry Representative.

Photocell eyes shall be aimed north.

635.19.04 Street name Signs – Street name signs shall be bolted directly to the pole arms in accordance with SS Drawing SP635-3.2.1.

635.19.05 Audible Signals – Audible Signals shall be installed on pedestrian signal heads in accordance with SS Drawing SP635-2.3.9. The Contractor shall aim and adjust the audible signal heads to the satisfaction of the Ministry Representative.

635.19.06 Emergency Vehicle Pre-emption Equipment – Emergency vehicle pre-emption equipment shall be installed in accordance with the Drawings and manufacturer's instructions. The Contractor shall undertake all commissioning and set-up of the equipment under the direction of the supplier's technical representative.

635.19.07 Small Overhead Signs – Small overhead signs are classed as signs 1200 mm x 900 mm or smaller. Small overhead signs shall be installed in accordance with SS Drawings SP635-3.2.2 through SP635-3.2.6.

Sign installations shall meet the requirements of SS 635.34.

635.19.08 Video Detection Equipment – Video Detection equipment shall be installed in accordance with the Drawings and manufacturer's instructions. The cameras shall be installed on signal arms or on special combination Type 2A luminaire / camera arms in accordance with SS Drawings SP635-2.3.12 and SP635-2.3.13. The Contractor shall undertake all commissioning and set-up of the equipment under the direction of the supplier technical representative.

635.19.09 Payment – Payment for the installation of each SIGNAL POLE will be at the Lump Sum Price for that pole.

The Lump Sum Price shall include all costs for the installation of traffic signal poles and arms, traffic signal heads including mounting hardware, pedestrian pushbuttons and signs, luminaires and photocells, audible signals, street name signs, small overhead signs, and fire indication lights; supply and installation of wiring to the handhole at the base of the pole, fuse holders, fuses and splices; and all other labour, equipment and material necessary to complete the installation.

635.19.10 Payment – Payment for the installation of LUMINAIRE POLES will be at the Unit Price for each pole.

The Unit Price shall include all costs for the installation of luminaire poles, luminaires and photocells, and frangible or breakaway bases; supply and installation of wiring to the handhole at the base of the pole, fuse holders, fuses and splices; and all other labour, equipment and materials necessary to complete the installation.

635.19.11 Payment – Payment for the supply and installation of EMERGENCY VEHICLE PRE-EMPTION EQUIPMENT will be made at the Lump Sum Price for each signal.

The Lump Sum Price shall include all costs for the installation of the pre-emption equipment, commissioning, set-up and adjusting.

635.19.12 Payment – Payment for the supply and installation of VIDEO DETECTION EQUIPMENT will be made at the Lump Sum Price for each signal.

The Lump Sum Price shall include all costs for the supply and installation of the video detection equipment, commissioning, set-up and adjusting.

635.20 Service Equipment – Service equipment shall be installed in accordance with SS Drawings SP635-2.4.1 through SP635-2.4.20.

Service Equipment shall include electrical panels and telephone demarcation panels.

Specialized service equipment (e.g., service kiosks) shall be installed in accordance with the Drawings and Special Provisions.

Service equipment shall be securely attached to the poles.

635.20.01 Payment – Payment for the installation of SERVICE EQUIPMENT will be at the Lump Sum Price for that service area.

The Lump Sum Price shall include all costs for the supply and installation of electrical service, distribution and disconnect panels and telephone demarcation panels including the pole where a separate service pole is required; supply and installation of service wiring including wiring between the disconnect and distribution panels; supply and